IEEE 754 Decimals for C++

The Boost.Decimal Library

MATT BORLAND

Start with a Quiz

What is the output of this program?

```cpp
#include <iomanip>
#include <iostream>

int main() {
  std::cout << std::setprecision(17) << 0.1 + 0.2;
}
```

0.30000000000000004 and why is this?

Where the extra came from

The bitwise representation of `float f = 0.1f;`

$$(-1)^0 \times 1.10011001100110011001101_2 \times 2^{123-127} = \frac{13421773}{2^{27}} = 0.100000001490116119384765625$$

The Reason? Five does not divide two

A fraction terminates in base b only if every prime factor of its denominator divides b.[1]

$$\frac{1}{10} \qquad 10 = 2 \times 5 \qquad 5 \nmid 2$$

No number of bits fixes this. The problem is not precision, but **representability**.

[1] Hardy & Wright, An Introduction to the Theory of Numbers, §9.2–9.3.

The solution is not "more precision"

These numerical types we are about to discuss are useful when these errors are unacceptable.

— Canonical example and many users are in the financial sector
— Human-entered data and databases
— Legal and regulatory compliance

A brief history and availability

— **IEEE 754-2008** specifies decimal floating point with two encodings and three interchange formats

— **ISO/IEC TR 24733** (2011) sketches a C++ binding. Never adopted into the standard.

— **N3871** (2014) second attempt at adding decimal types to the C++ standard

— **C23** adds `_Decimal32` `_Decimal64` `_Decimal128` as an optional feature

— **GCC** libstdc++ ships these types on selected targets (`<decimal/decimal>`), without an accompanying standard library

— **IBM's libdfp** fills the library gap

— **Intel** ships a library with its own types

Boost.Decimal

- Header-only, **no dependencies**, C++14
- Portable: tested on `x86_64`, `x86_32`, ARM64, ARM32, s390x; emulated PPC64LE and Cortex-M
- Complete Standard Library: `<cmath>`, `<charconv>`, `<format>`, etc.
- 3rd Party Library Integration: Boost.Math, {fmt}
- `constexpr` throughout
- CUDA support for the types and much of the math

This talk will be divided into five parts

1. Inside the bits - how decimal floating point is represented
2. The types - an overview of the different types provided by the library, and how to use them
3. The Standard Library - similarities and differences with the STL
4. Performance - Comparisons between Boost.Decimal, libstdc++, and Intel decimal types
5. When to reach for it - Usecases from known users and takeaways

Part 1 of 5

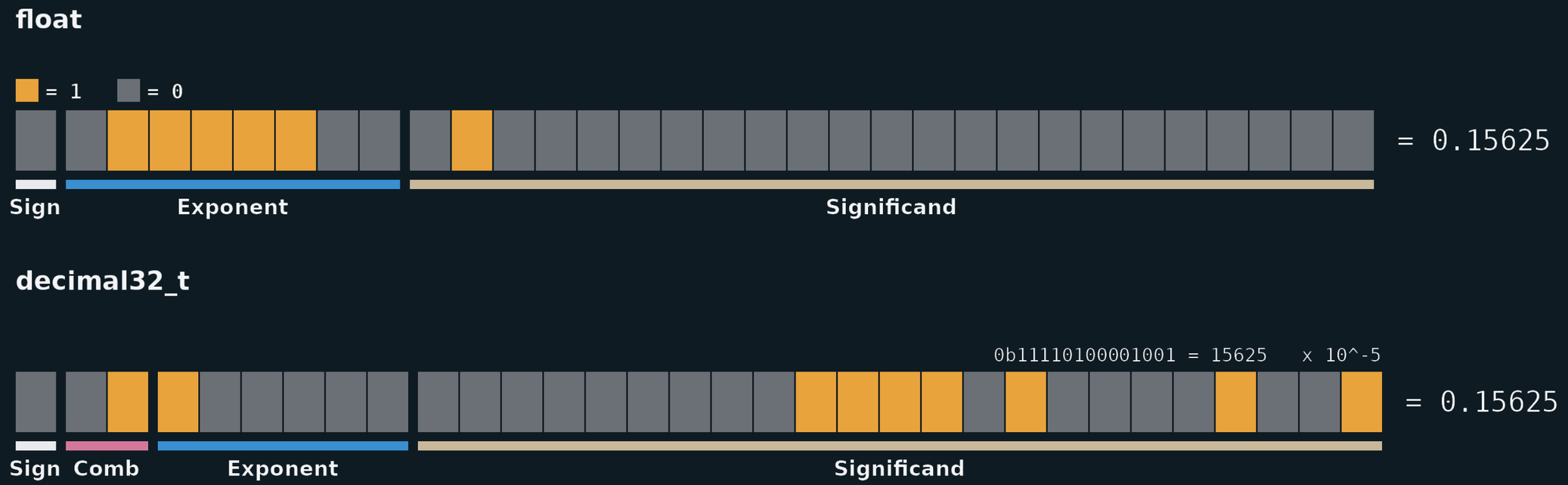
1. **Inside the bits**
2. The Types
3. The Standard Library
4. Performance
5. When to reach for it

A Binary Floating Point Refresher

`float` bits: 1 sign + 8 exponent + 23 significand = 32.

$$(-1)^{0} \times 2^{124-127} \times 1.01_{2} = 1.25 \times 2^{-3} = 1.25 \div 8 = 0.15625$$

Same value with one additional field.



$$15625 \times 10^{-5} = 0.15625$$

Why the extra field?

decimal32_t holds 7 decimal digits -> max significand 9,999,999

2^23 = 8,388,608 < 9,999,999
2^24 = 16,777,216

You need 24 bits, but sign and exponent leave you with only 23.

Four cases

steer	layout	sig. bits
00	s | 00 | eeeeeeee | ttttttttttttttttttttttt	23
01	s | 01 | eeeeeeee | ttttttttttttttttttttttt	23
10	s | 10 | eeeeeeee | ttttttttttttttttttttttt	23
11	s | 11 | eeeeeeee | [100] + ttttttttttttttttttttt	24

100 + 21 trailing 1s = 10,485,759 > 9,999,999

Watch the fields move for adjacent integers

decimal32_t{8388607} steering != 11

decimal32_t{8388608} steering == 11

0.1, both ways

float

decimal32_t

BID and DPD

IEEE 754 specifies two encodings:

— **BID**: Binary Integer Significand. Intended for software implementations.

— **DPD**: Densely Packed Decimal. Intended for hardware implementations.

Boost.Decimal uses **BID**, and can convert either way:

```
constexpr std::uint32_t to_bid_d32(decimal32_t val)      noexcept;
constexpr decimal32_t   from_bid_d32(std::uint32_t bits) noexcept;
constexpr std::uint32_t to_dpd_d32(decimal32_t val)      noexcept;
constexpr decimal32_t   from_dpd_d32(std::uint32_t bits) noexcept;
```

BID vs DPD for our adjacent integers

	BID	DPD
`decimal32_t{8388607}`	`0x32FFFFFF`	`0x6A573B07`
`decimal32_t{8388608}`	`0x6CA00000`	`0x6A573B08`

BID rebuilds the word.

DPD increments by one.

Now the part with no binary analogue

One value. Seven encodings.

The set of representations a floating-point number maps to is called the floating-point number's cohort.
— IEEE 754-2008

```
decimal32_t{      3,  2}   ->   0x33800003
decimal32_t{     30,  1}   ->   0x3300001E
decimal32_t{    300,  0}   ->   0x3280012C
decimal32_t{3000000, -4}   ->   0x30ADC6C0
```

— All compare equal with `operator==`

— All **hash equal**

— None are bitwise equal

Normalizing

A value is **normalized** when the cohort effect is removed by appending zeros to the significand and reducing the exponent until it carries the type's full precision:

$$3 \times 10^{2} \quad \longrightarrow \quad 3000000 \times 10^{-4}$$

Every arithmetic operation and every comparison has to handle the effects of cohorts.

This is among the most expensive parts of these operations.

Part 2 of 5

1. Inside the bits
2. **The Types**
3. The Standard Library
4. Performance
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One Include

```
#include <boost/decimal.hpp>
```

Header-only. No dependencies. C++14.

```
import boost.decimal; // C++20 module
```

— Available individually through vcpkg and conan, or with Boost in most package managers

— Build with b2 or CMake

Six types

Three IEEE 754 compliant types:

	bytes	digits10	exponent range
decimal32_t	4	7	-95 .. +96
decimal64_t	8	16	-383 .. +384
decimal128_t	16	34	-6143 .. +6144

Three performance-focused types:

decimal_fast32_t	8	7	-95 .. +96
decimal_fast64_t	16	16	-383 .. +384
decimal_fast128_t	32	34	-6143 .. +6144

The performance costs exactly double the size.

What the extra bytes buy

```
class decimal32_t final {
    std::uint32_t bits_;              // IEEE 754 BID
};

class decimal_fast32_t final {
    std::uint32_t significand_;       // stored decoded and normalized
    std::uint8_t  exponent_;
    bool          sign_;
};
```

Which one to use?

Does the bit pattern matter?

— **Yes**: a file, a socket, a column, another language, another vendor
→ `decimal32_t decimal64_t decimal128_t`

— **No**: computed, compared, and discarded
→ `decimal_fast32_t decimal_fast64_t decimal_fast128_t`

Construction

```
decimal32_t a {1, 1};                                  // 1e1
decimal32_t b {-2, -1};                                // -0.2
decimal32_t c {2U, -1, construction_sign::negative};   // -0.2
decimal32_t d {"4.3e-02"};                             // string or string_view
auto        e {"3.14159265358979"_DF};                 // literal, rounds to fit
```

It promotes like you expect

```
decimal32_t{}         + decimal64_t{}          ->  decimal64_t
decimal64_t{}         * 2                      ->  decimal64_t
decimal64_t{}         + decimal_fast32_t{}     ->  decimal64_t
decimal_fast128_t{} - decimal128_t{}           ->  decimal_fast128_t
```

Wider precision wins. On a tie, **fast wins**.

What it will not do quietly

```
decimal64_t d {3.14};          // explicit only, and discouraged
int n = d;                     // ill-formed
```

Part 3 of 5

1. Inside the bits
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3. **The Standard Library**
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What comes in the box

`<cmath>`	over a hundred functions, all constexpr
`<charconv>`	to_chars / from_chars, plus two decimal-only formats
`<format>`	and {fmt}
`<cstdlib>`	strtod32 / strtod64 / strtod128
`<cfenv>`	rounding modes
`<cfloat>`	evaluation modes
`<functional>`	std::hash, all six types
`<limits>`	numeric_limits, all six types
`<string>`	to_string, stod32 ...
`<iostream>`	operator<< and operator>>
`<numbers>`	pi, e, ln2, ... per type

Everything lives in `boost::decimal`, except literals, which live in `boost::decimal::literals`.

Generic code: the `std::swap` two-step

```cpp
template <typename T>
void sin_identity(T val)
{
    using std::sin;
    using boost::decimal::sin;

    sin(val);           // unqualified. float, double, decimal128_t — all fine.
}
```

All of it is constexpr

```
constexpr decimal64_t two  {2};
constexpr decimal64_t root {sqrt(two)};

static_assert(root == "1.414213562373095"_DD, "");
```

`std::sqrt` is not `constexpr` until C++26.

Two ways to decompose a value

Three spellings of `300`, taken apart two ways:

```
decimal32_t{300}          decompose  ->  sig 300,      exp  0
                          frexp10    ->  sig 3000000,  exp -4

decimal32_t{3, 2}         decompose  ->  sig 3,        exp  2
                          frexp10    ->  sig 3000000,  exp -4

decimal_fast32_t{3, 2}    decompose  ->  sig 3000000,  exp -4
                          frexp10    ->  sig 3000000,  exp -4
```

— `decompose` retains the cohort

— `frexp10` normalizes it away

— The fast types return the same for both since they store normalized

A deeper look into <charconv>

chars_format options

```
enum class chars_format : unsigned
{
    scientific,
    fixed,
    hex,
    cohort_preserving_scientific,     // not in <charconv>
    cohort_preserving_fixed,          // not in <charconv>
    general = fixed | scientific
};
```

The round trip

<charconv> using chars_format::cohort_preserving_scientific

```
"3e+02"          ->  0x33800003  ->  "3e+02"
"3.0e+02"        ->  0x3300001E  ->  "3.0e+02"
"3.00e+02"       ->  0x3280012C  ->  "3.00e+02"
"3.000e+02"      ->  0x32000BB8  ->  "3.000e+02"
"3.0000e+02"     ->  0x31807530  ->  "3.0000e+02"
"3.00000e+02"    ->  0x310493E0  ->  "3.00000e+02"
"3.000000e+02"   ->  0x30ADC6C0  ->  "3.000000e+02"
```

Sizing buffers

```
template <typename DecimalType, int Precision = std::numeric_limits<DecimalType>::max_digits10>
class formatting_limits
{
public:
    static constexpr std::size_t scientific_format_max_chars;
    static constexpr std::size_t fixed_format_max_chars;
    static constexpr std::size_t hex_format_max_chars;
    static constexpr std::size_t cohort_preserving_scientific_max_chars;
    static constexpr std::size_t cohort_preserving_fixed_max_chars;
    static constexpr std::size_t general_format_max_chars;
    static constexpr std::size_t max_chars;
};
```

This simplifies correct buffer sizing to:

```
char buf[formatting_limits<decimal64_t>::scientific_format_max_chars];
```

Integration with other libraries

Two ways to format

```
                    needs                                   standard
<fmt/format.h>      {fmt} 11 or newer                       C++14
<format>            GCC 13 · Clang 18 · MSVC 19.40          C++20

fmt::format("{:*>+12.2e}", val);      // [***+3.14e+00]
std::format("{:*>+12.2e}", val);      // [***+3.14e+00]
```

One extra formatting specifier

```
g G    general
e E    scientific
f      fixed
x X    hex
a A    cohort preserving scientific
```

```
fmt::format("{}",   d);      // 300
fmt::format("{:a}", d);      // 3.00e+02
```

Rounding

Binary Floating Point Rounding

In <cfenv> we have the following four macros useable with `std::fesetround()`:

1. `FE_DOWNWARD` - Rounding towards negative infinity
2. `FE_TONEAREST` - Rounding towards nearest representable value
3. `FE_TOWARDZERO`- Rounding towards zero
4. `FE_UPWARD` - Rounding towards positive infinity

The Five Decimal Rounding Modes

All modes are in `enum class rounding_mode` and useable with `boost::decimal::fesetround()`:

1. `fe_dec_downward`
2. `fe_dec_to_nearest` - To nearest, ties to even
3. `fe_dec_to_nearest_from_zero`
4. `fe_dec_toward_zero` - The opposite of 3
5. `fe_dec_upward`

Default is #2, per IEEE 754 4.3.3 and is called "banker's rounding", and is available also as `fe_dec_default`

Example of Rounding Mode

```cpp
#include <boost/decimal.hpp>
#include <iostream>

int main() {
    using namespace boost::decimal::literals;
    using boost::decimal::decimal32_t;

    boost::decimal::fesetround(boost::decimal::rounding_mode::fe_dec_upward); // NOT THREAD-SAFE

    const decimal32_t lhs {"5e+50"_DF};
    const decimal32_t rhs {"4e+40"_DF};
    const decimal32_t sum {lhs + rhs};

    std::cout << "5e50 + 4e40 = " << sum << std::end;
}
```

Output: 5e50 + 4e40 = 5.000001e+50

Compile Time Rounding

Very similar to the run-time `enum class`, but we now have the following macros:

1. `BOOST_DECIMAL_FE_DEC_DOWNWARD`
2. `BOOST_DECIMAL_FE_DEC_TO_NEAREST`
3. `BOOST_DECIMAL_FE_DEC_TO_NEAREST_FROM_ZERO`
4. `BOOST_DECIMAL_FE_DEC_TOWARD_ZERO`
5. `BOOST_DECIMAL_FE_DEC_UPWARD`

Changing the Compile Time Rounding Mode

```cpp
#define BOOST_DECIMAL_FE_DEC_DOWNWARD   // before ANY decimal header

#include <boost/decimal/decimal32_t.hpp>
#include <boost/decimal/literals.hpp>

using namespace boost::decimal::literals;
using boost::decimal::decimal32_t;

constexpr decimal32_t lhs {"5e+50"_DF};
constexpr decimal32_t rhs {"4e+40"_DF};
constexpr decimal32_t res {lhs - rhs};

static_assert(res == "4.999999e+50"_DF, "Incorrectly rounded result");
```

Boost.Math Integration

Bollinger bands over a year of AAPL closes.

```cpp
#include <boost/math/statistics/univariate_statistics.hpp>
#include <boost/decimal.hpp>

std::vector<decimal64_t> closes;

// Import closing data from a data source of AAPL

const decimal64_t mean   = boost::math::statistics::mean(closes);
const decimal64_t median = boost::math::statistics::median(closes);
const decimal64_t var    = boost::math::statistics::variance(closes);
const decimal64_t sigma  = boost::decimal::sqrt(var);
```

```
  Mean Closing Price: $207.20
  Standard Deviation: $25.45
Upper Bollinger Band: $258.11
Lower Bollinger Band: $156.30
```

On the GPU

```
#define BOOST_DECIMAL_ENABLE_CUDA
#include <boost/decimal/decimal64_t.hpp>

__global__ void add(const decimal64_t* a, const decimal64_t* b,
                    decimal64_t* out, int n)
{
    const int i {blockDim.x * blockIdx.x + threadIdx.x};
    if (i < n) { out[i] = a[i] + b[i]; }
}
```

Device results compare to the same loop run on the host

Debugging Pretty Printers

decimal32_t - cohort preserving

```
> max = {const boost::decimal::decimal32_t} 9.999999e+96
> min = {const boost::decimal::decimal32_t} 1e-95
> short_num = {const boost::decimal::decimal32_t} 3.140e+00
> pos_inf = {const boost::decimal::decimal32_t} INF
> neg_inf = {const boost::decimal::decimal32_t} -INF
> qnan = {const boost::decimal::decimal32_t} QNAN
> snan = {const boost::decimal::decimal32_t} SNAN
> payload_nan = {const boost::decimal::decimal32_t} QNAN(7)
```

decimal_fast32_t - always normalized

```
> max = {const boost::decimal::decimal_fast32_t} 9.999999e+96
> min = {const boost::decimal::decimal_fast32_t} 1.000000e-95
> short_num = {const boost::decimal::decimal_fast32_t} 3.140000e+00
> pos_inf = {const boost::decimal::decimal_fast32_t} INF
> neg_inf = {const boost::decimal::decimal_fast32_t} -INF
> qnan = {const boost::decimal::decimal_fast32_t} QNAN
> snan = {const boost::decimal::decimal_fast32_t} SNAN
> payload_nan = {const boost::decimal::decimal_fast32_t} QNAN(7)
```

Part 4 of 5

1. Inside the bits
2. The Types
3. The Standard Library
4. **Performance**
5. When to reach for it

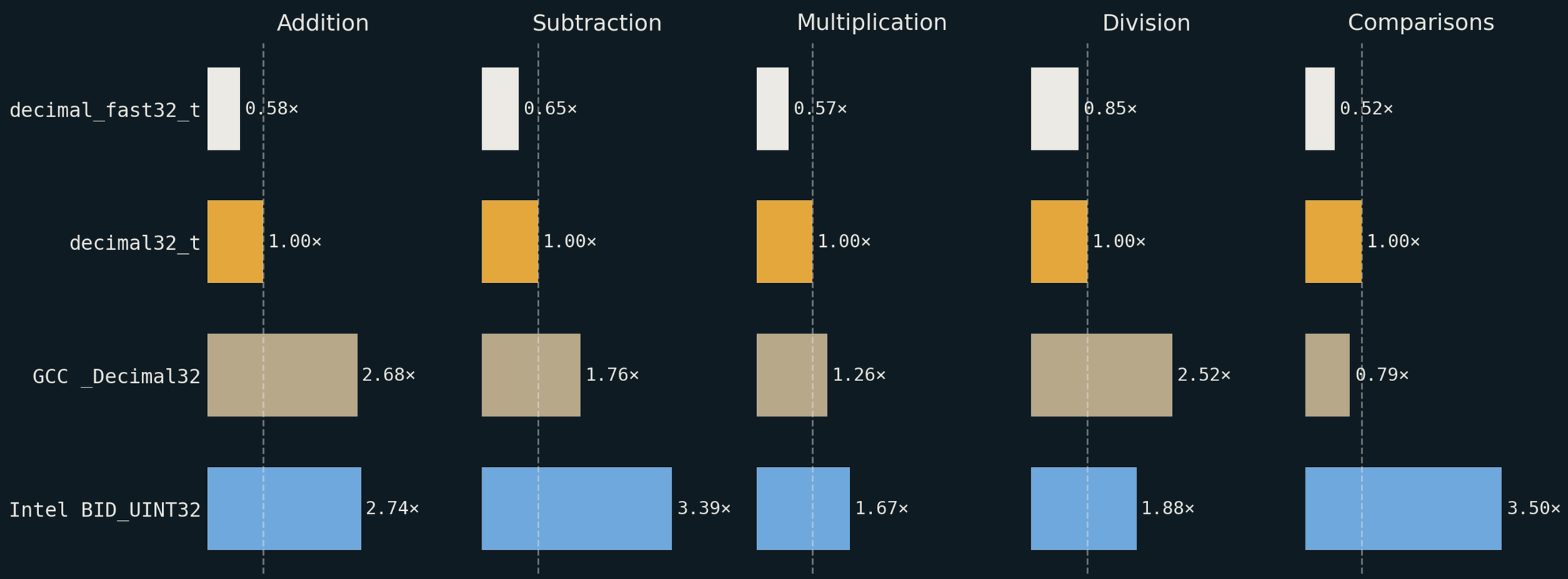
Benchmarks

Complete Benchmarks are available for the following systems:

1. x86_64 Linux
2. x86_32 Linux
3. x86_64 Windows
4. ARM64 Windows
5. ARM64 MacOS

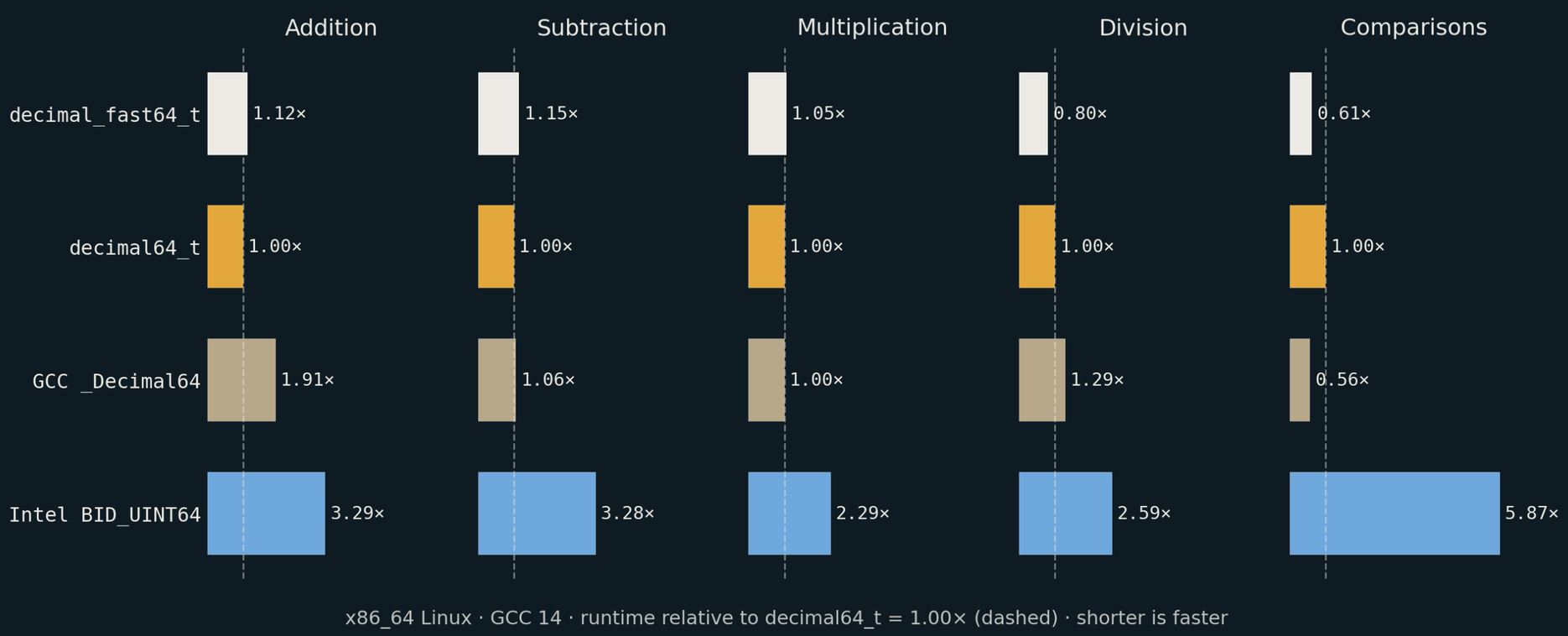
GCC `_DecimalXX` and Intel `BID_UINTXX` are included where available for completeness

32-bit Types



x86_64 Linux · GCC 14 · runtime relative to decimal32_t = 1.00× (dashed) · shorter is faster

64-bit Types



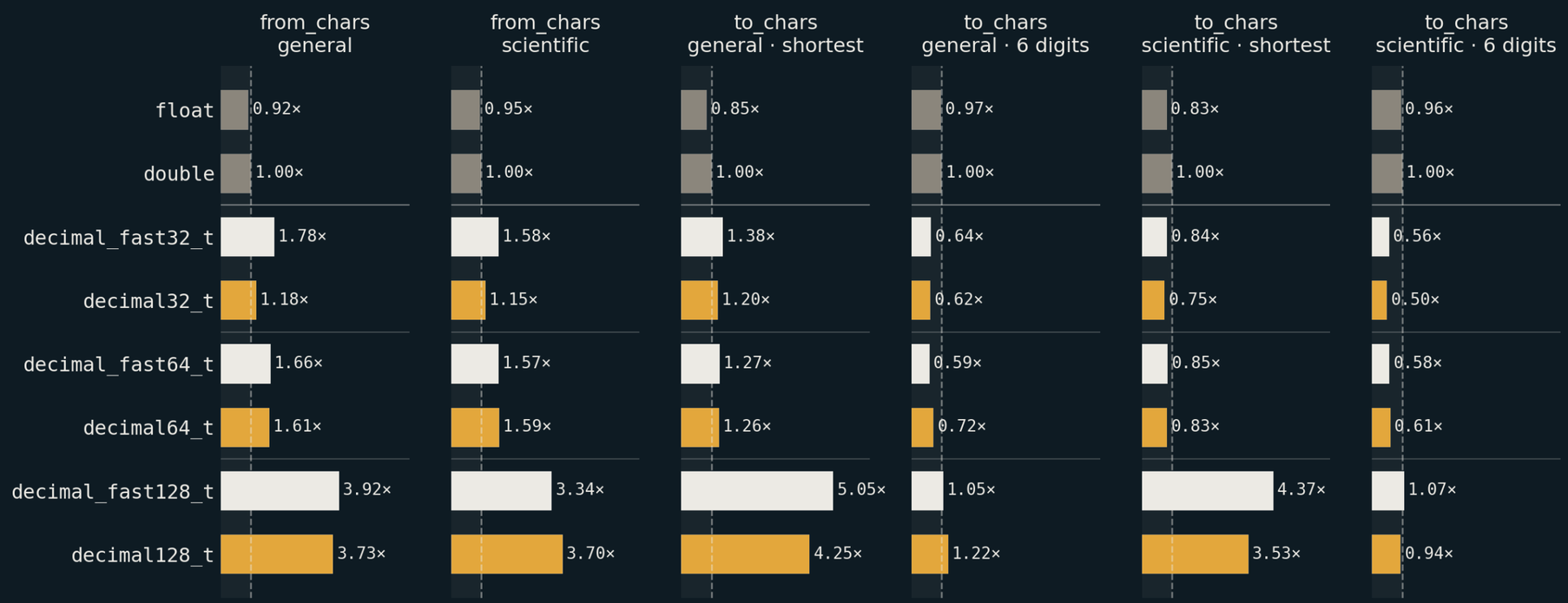
One honest loss: GCC `_Decimal64` compares faster than we do

128-bit Types

x86_64 Linux · GCC 14 · runtime relative to decimal128_t = 1.00× (dashed) · shorter is faster

The closest race of the three widths

Where decimal beats the hardware



x86_64 Linux · runtime relative to double = 1.00× (dashed) · left of the line beats the hardware type

— <charconv> is software for every type so the hardware advantage disappears

Part 5 of 5

1. Inside the bits
2. The Types
3. The Standard Library
4. Performance
5. **When to reach for it**

Use Cases Revisited

At the top of the presentation we suggested:

— Canonical example and many users are in the financial sector

— Human-entered data and databases

— Legal and regulatory compliance

Current Clients Applications

— A trading firm, because their home-brew fixed point could not represent the smallest divisible unit of a Bitcoin (A Satoshi = 1/100,000,000 of a Bitcoin)

— Several quant firms, who use `<charconv>` extensively

— A database firm, who needed `decimal128_t` to be run on ARM

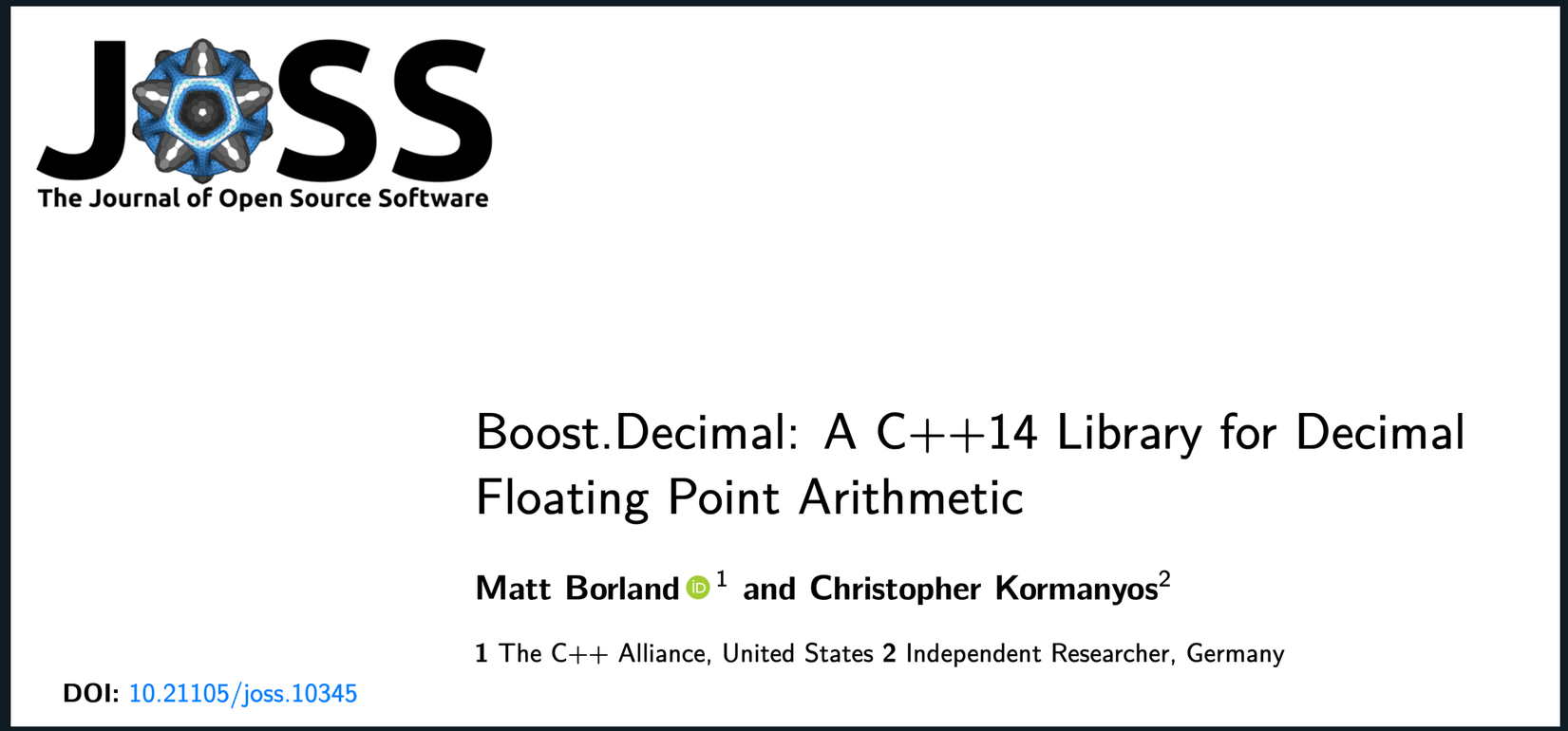
— Boost.MySQL for the DECIMAL data type (in conjunction with Boost.Multiprecision)

Takeaways

— Decimal Floating Point types can provide a new way to handle data and computations, but there is a performance cost

— Boost.Decimal is portable, performant, and available everywhere.

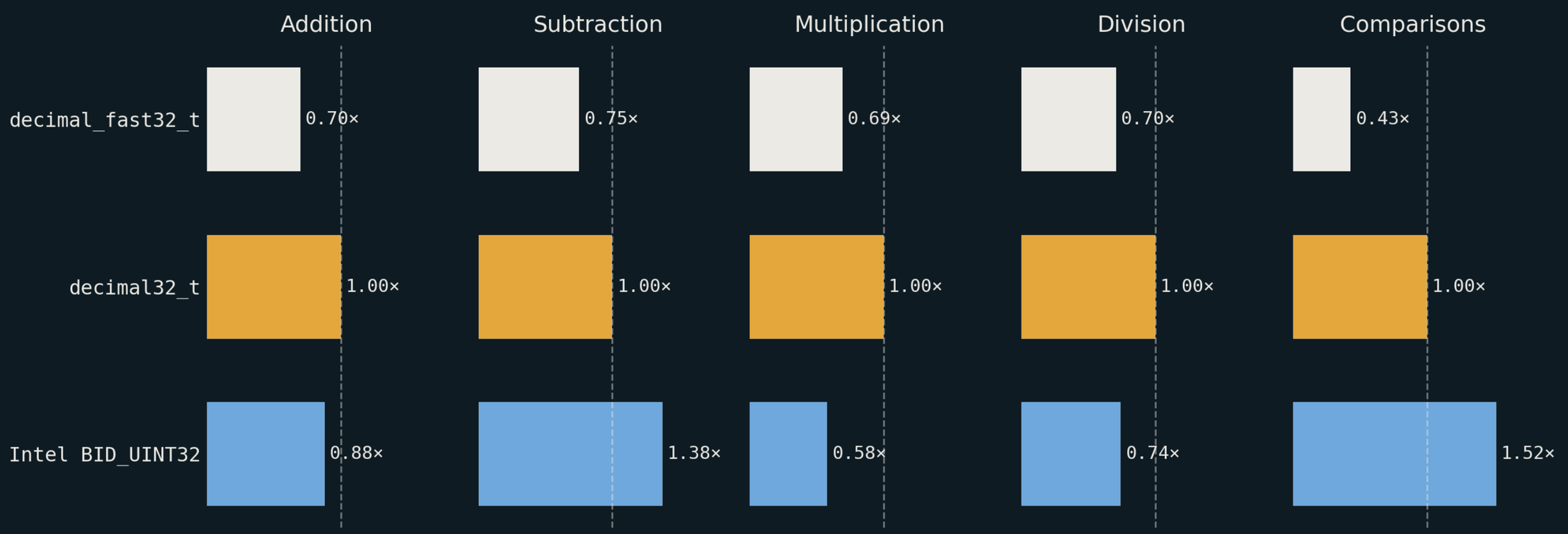
Questions and Citation



https://joss.theoj.org/papers/10.21105/joss.10345 or CITATION.cff in the repo

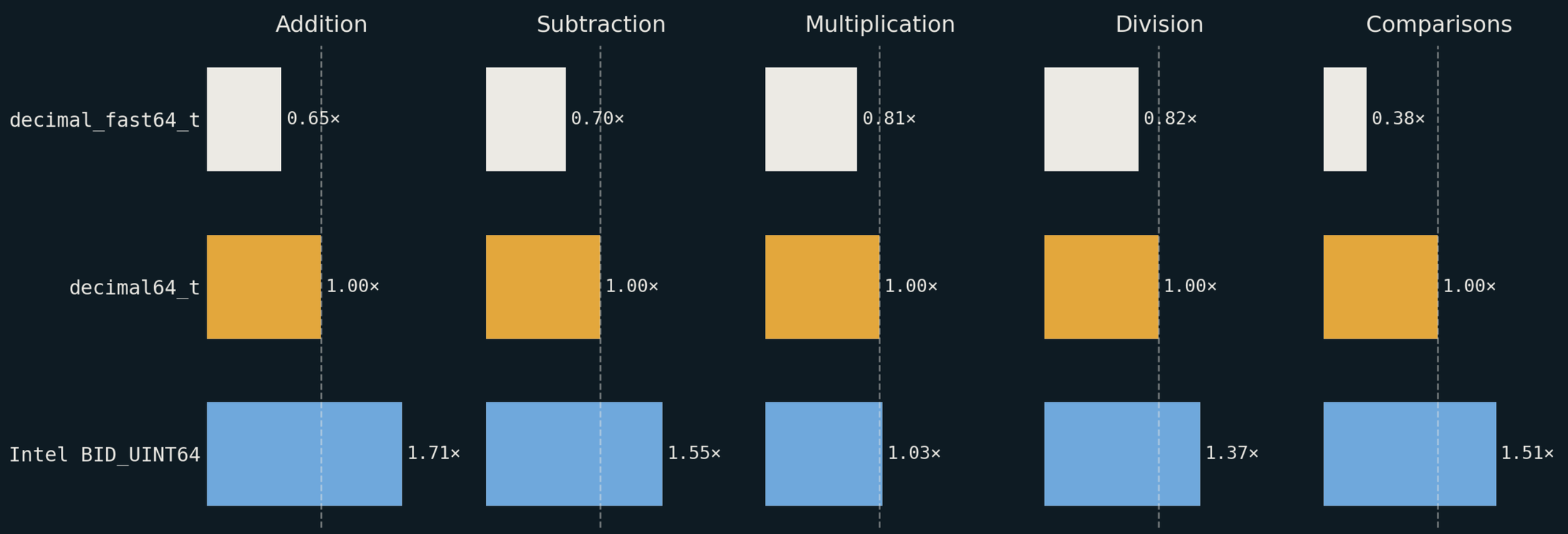
Appendix

x86_64 Linux · Intel oneAPI — 32 bits



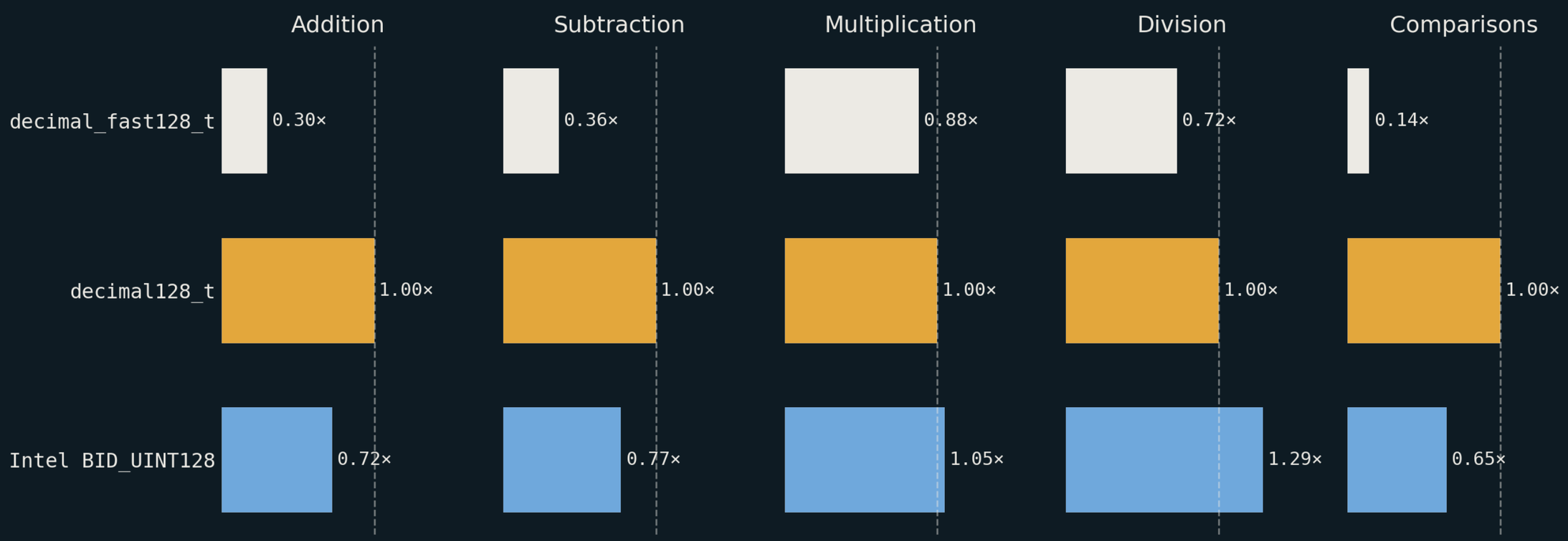
x86_64 Linux · Intel oneAPI (icpx / icx) · runtime relative to decimal32_t = 1.00× (dashed) · shorter is faster

x86_64 Linux · Intel oneAPI — 64 bits



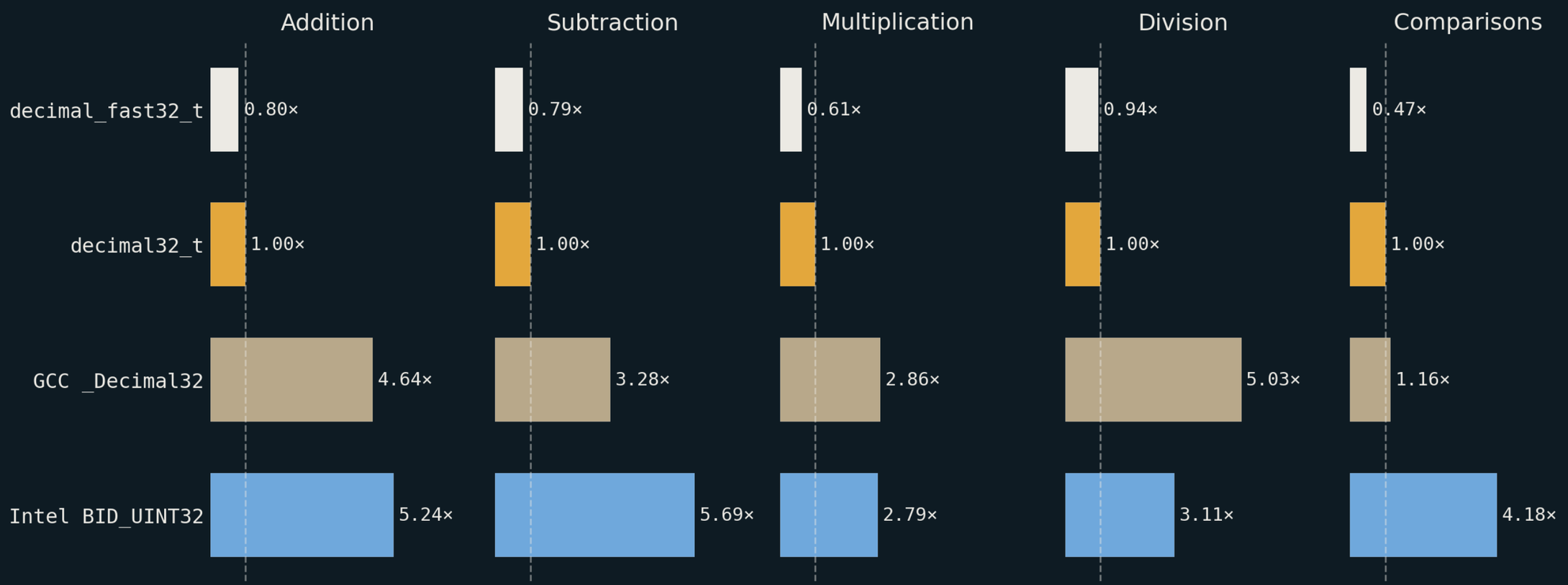
x86_64 Linux · Intel oneAPI (icpx / icx) · runtime relative to decimal64_t = 1.00× (dashed) · shorter is faster

x86_64 Linux · Intel oneAPI — 128 bits



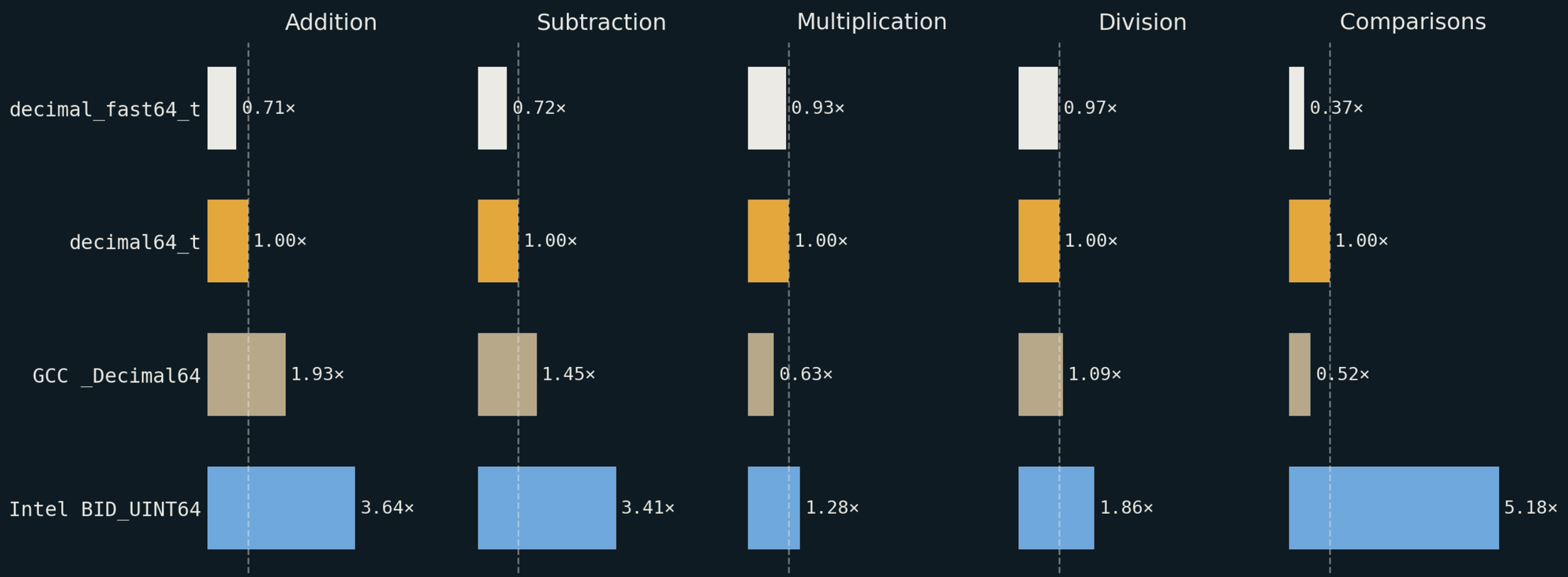
x86_64 Linux · Intel oneAPI (icpx / icx) · runtime relative to decimal128_t = 1.00× (dashed) · shorter is faster

x86_32 Linux · GCC 14 (-m32) — 32 bits



x86_32 Linux · GCC 14 · -m32 · runtime relative to decimal32_t = 1.00× (dashed) · shorter is faster

x86_32 Linux · GCC 14 (-m32) — 64 bits

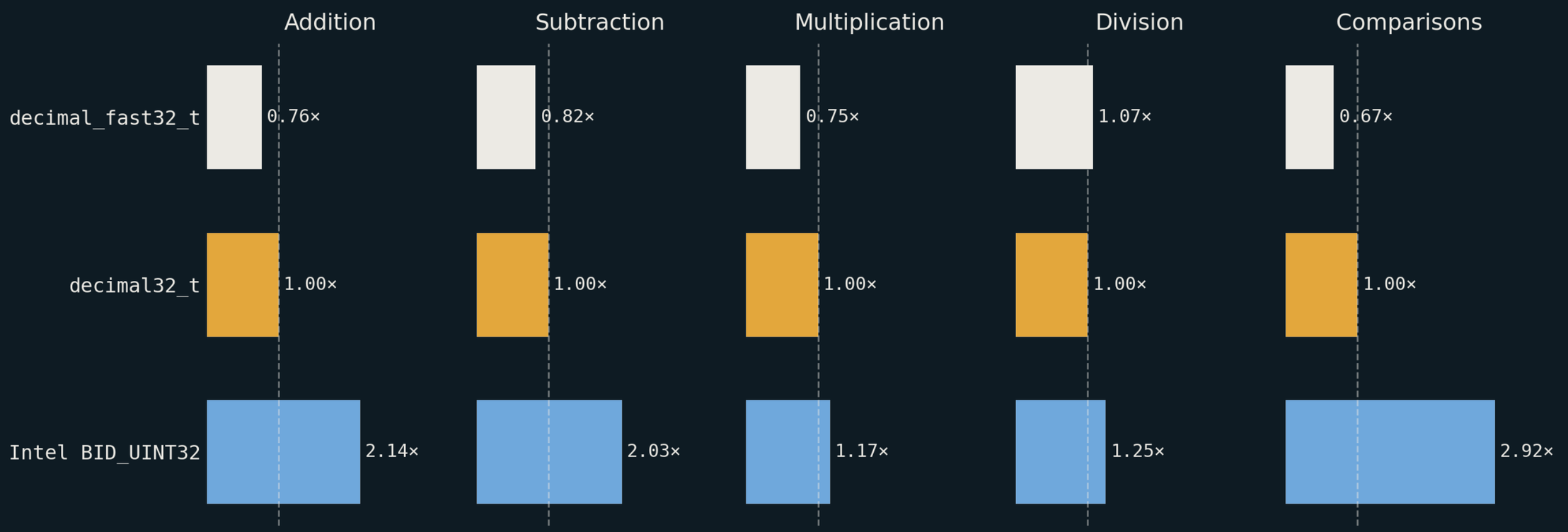


x86_32 Linux · GCC 14 · -m32 · runtime relative to decimal64_t = 1.00× (dashed) · shorter is faster

x86_32 Linux · GCC 14 (-m32) — 128 bits

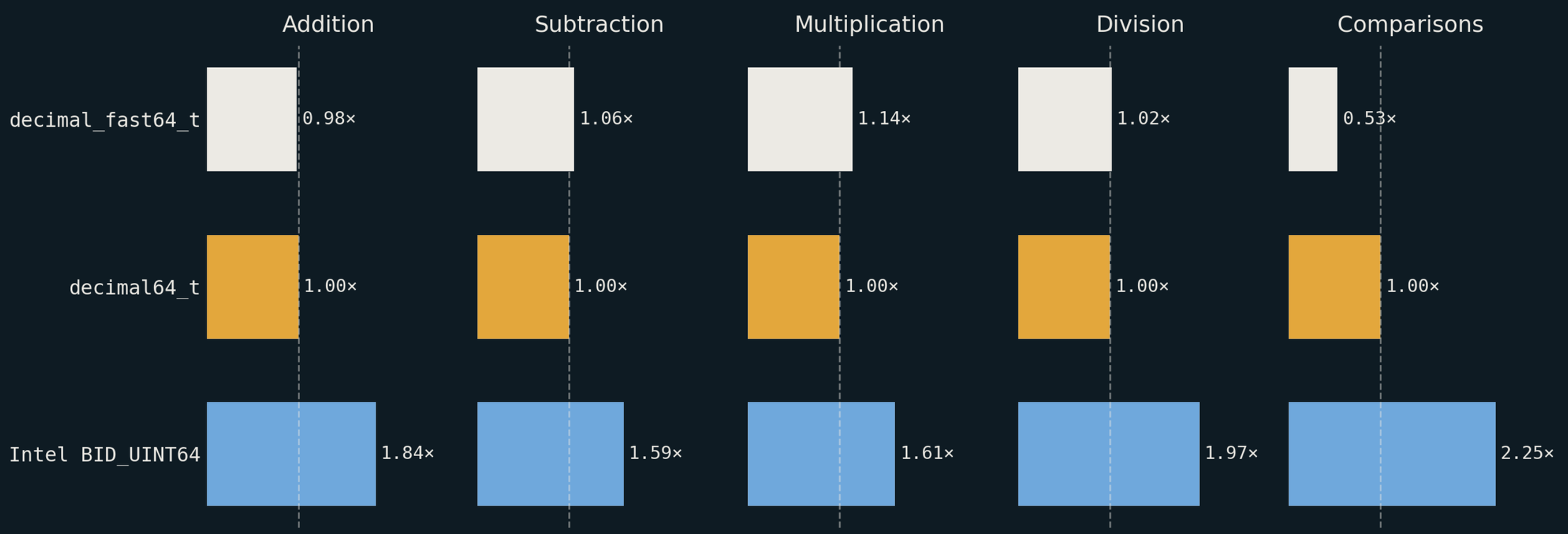
x86_32 Linux · GCC 14 · -m32 · runtime relative to decimal128_t = 1.00× (dashed) · shorter is faster

x86_64 Windows · MSVC — 32 bits



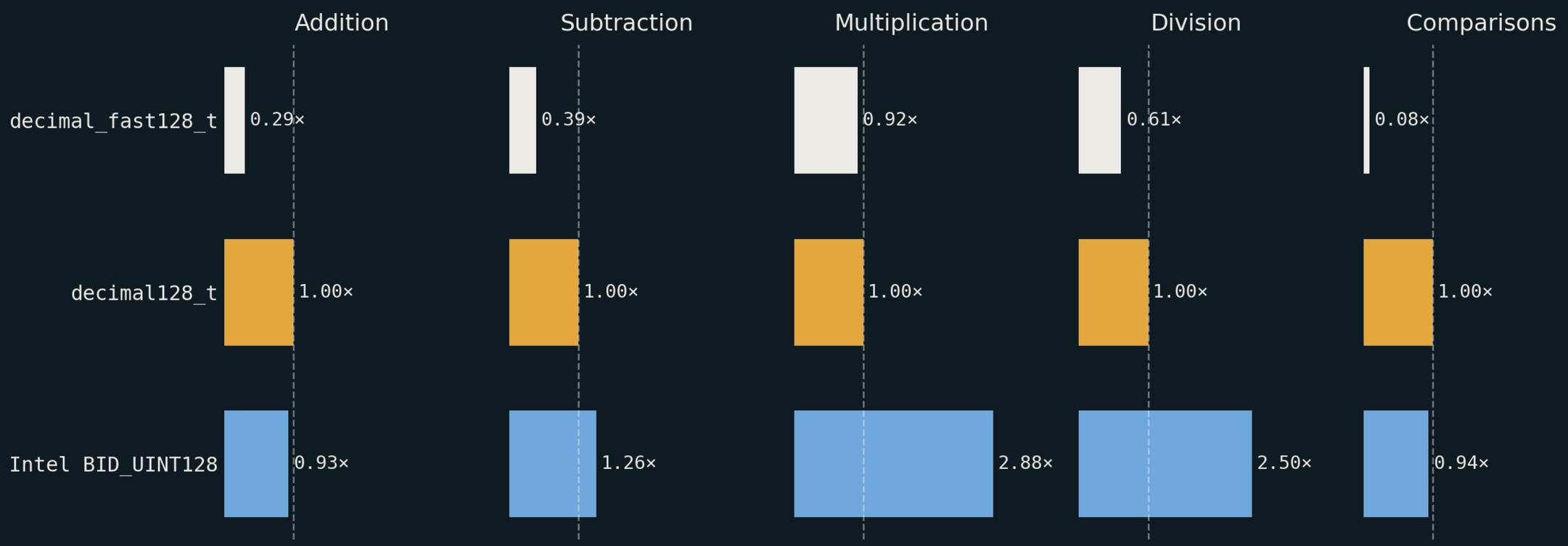
x86_64 Windows · MSVC · runtime relative to decimal32_t = 1.00× (dashed) · shorter is faster

x86_64 Windows · MSVC — 64 bits



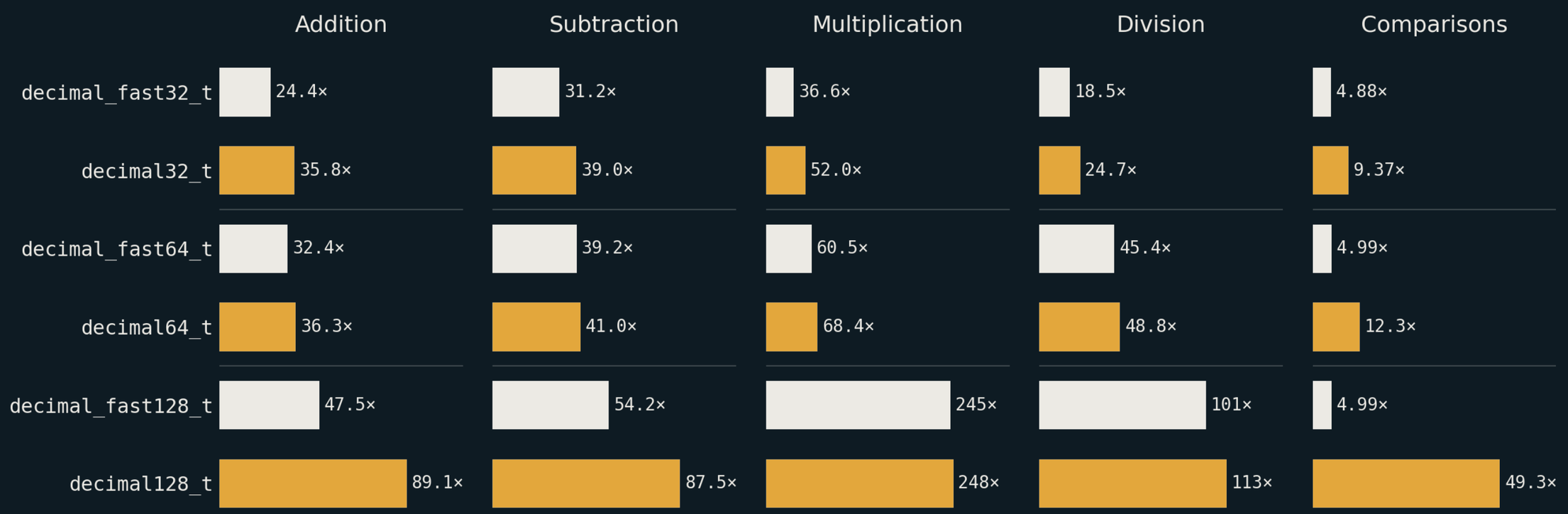
x86_64 Windows · MSVC · runtime relative to decimal64_t = 1.00× (dashed) · shorter is faster

x86_64 Windows · MSVC — 128 bits



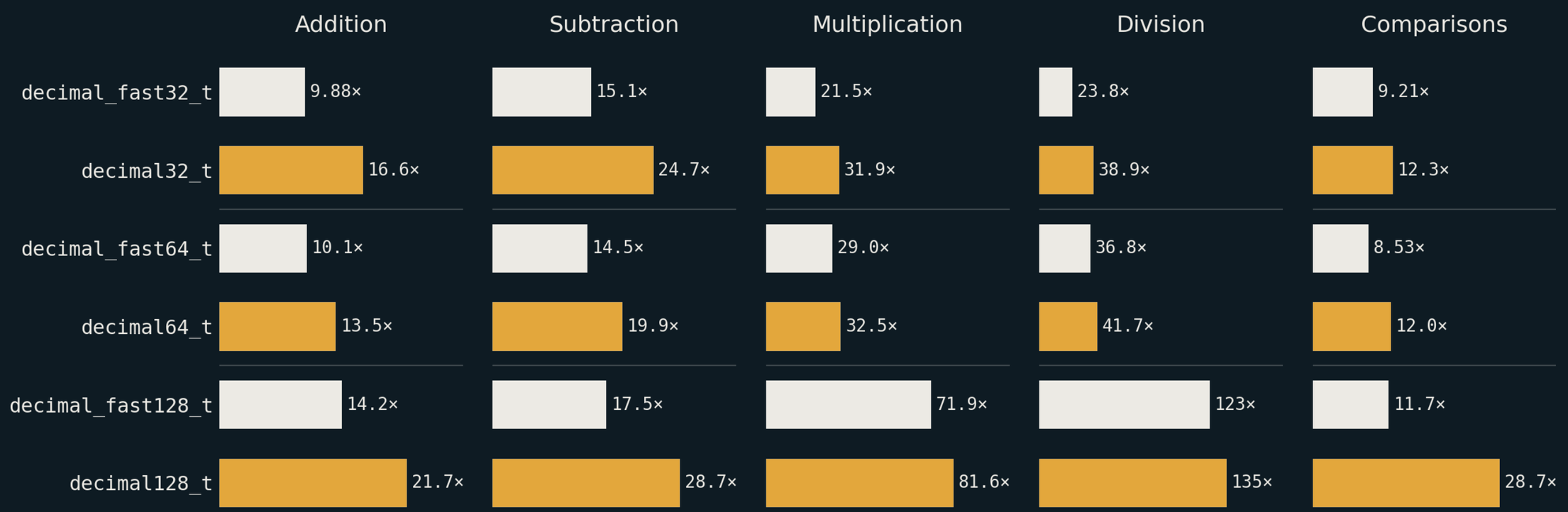
x86_64 Windows · MSVC · runtime relative to decimal128_t = 1.00× (dashed) · shorter is faster

ARM64 Windows · MSVC



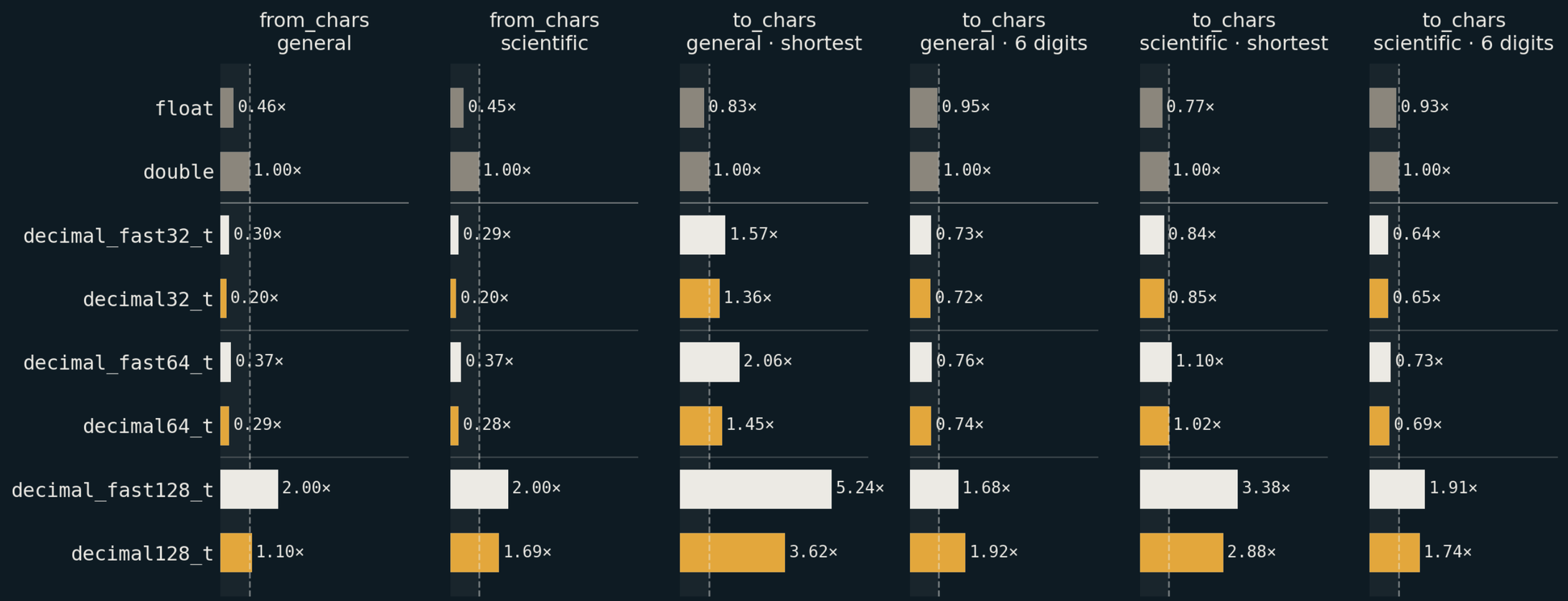
ARM64 Windows · MSVC · runtime relative to hardware double = 1.00× · shorter is faster · panels scaled independently

ARM64 macOS · M4 Max



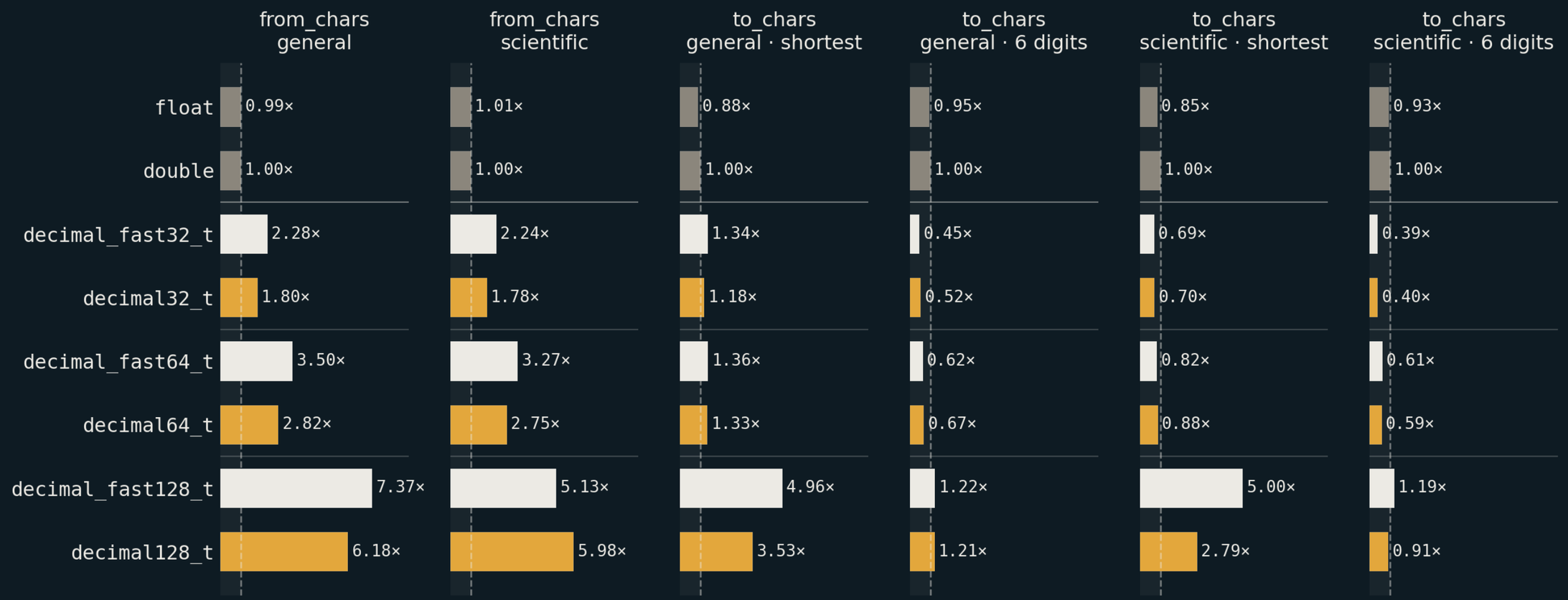
ARM64 macOS · M4 Max · Clang 22 · runtime relative to hardware double = 1.00× · shorter is faster · panels scaled independently

<charconv> — x86_64 Windows · MSVC



x86_64 Windows · MSVC · runtime relative to double = 1.00× (dashed) · left of the line beats the hardware type

<charconv> — ARM64 macOS



ARM64 macOS · M4 Max · Clang 22 · runtime relative to double = 1.00× (dashed) · left of the line beats the hardware type